Universität Stuttgart
Institute of Biomedical Engineering

BMT

Models and Test Methods in Biomedical Engineering

Summer semester 2026

Practice – Test plan preparation + Test setup

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Practice schedule

1	KW20	• Assignment description Hydrogel model • Renal* artery segmentation (with 3D Slicer) • Mold design (with Siemens NX)	May 12
2	KW21	• Continuation of mold design (with Siemens NX) • 3D printing preparation (with Preform)	May 19
	KW22	3D printing job to be sent per Email to Tobias Mörschel	May 28 at 5 pm
	KW23	Mold leak test	June 2
3	KW24	Test setup • Test plan preparation • Requirements definition • Outline of initial design concepts	June 9
4	KW25	• Short presentation (5 min): Test plan and design concepts • Test setup design finalization and 3D printing (with Bambulab)	June 16 **Deadline: June 19 at 5 pm**
5	KW26	• Model manufacturing • Assembly, initial operation, qualification	June 23
6	KW27	Testing and report	June 30
7	KW29	Team presentations	July 14

Flexibility and structural stability

User requirements (safety):

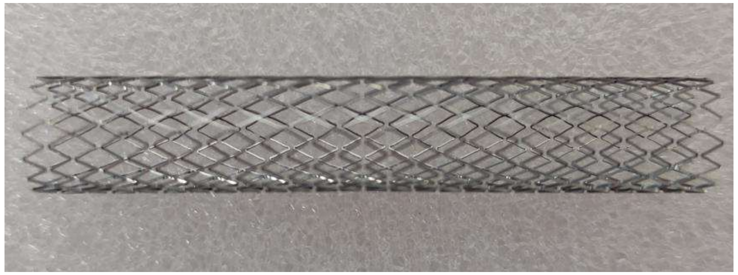
- Carotid stent must not disturb the hemodynamics
- Carotid stent must not damage the vessel while bending in the common carotid artery.

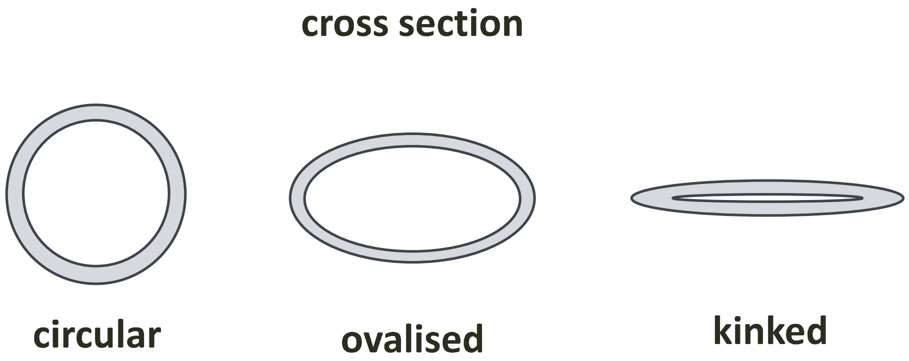
Engineering specification:

- Flexibility: Ability to **yield** under **bending force**
- Structural stability: Ability to undergo bending **without** cross-sectional **deformation (ovalization or kinking)**

Ovalization: When a circular or cylindrical structure bends, its circular cross-section deforms into an oval shape.

Kinking: sharp, localized bend





In vitro flexibility and structural stability test of carotid stents

Objective

To compare **two generations of stents** with regard to the **engineering specifications** by:

- Determining the **force** needed to bend the stent
- Determining the starting point of **ovalization**

in **two** different **conditions**: fully-expanded vs. implanted in a hydrogel model

Test material

- Nitinol stents, laser-cut with an 8 mm diameter and 50 mm length
    - Generation 1: rigid
    - Generation 2: flexible

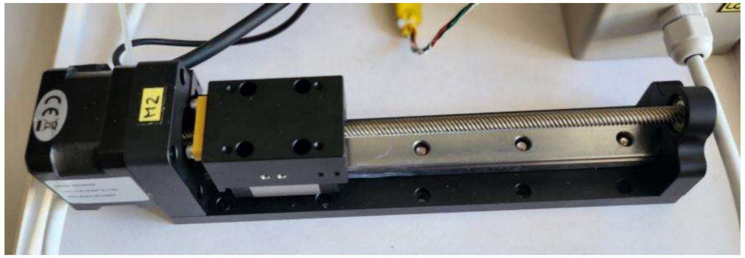
Acceptance criteria

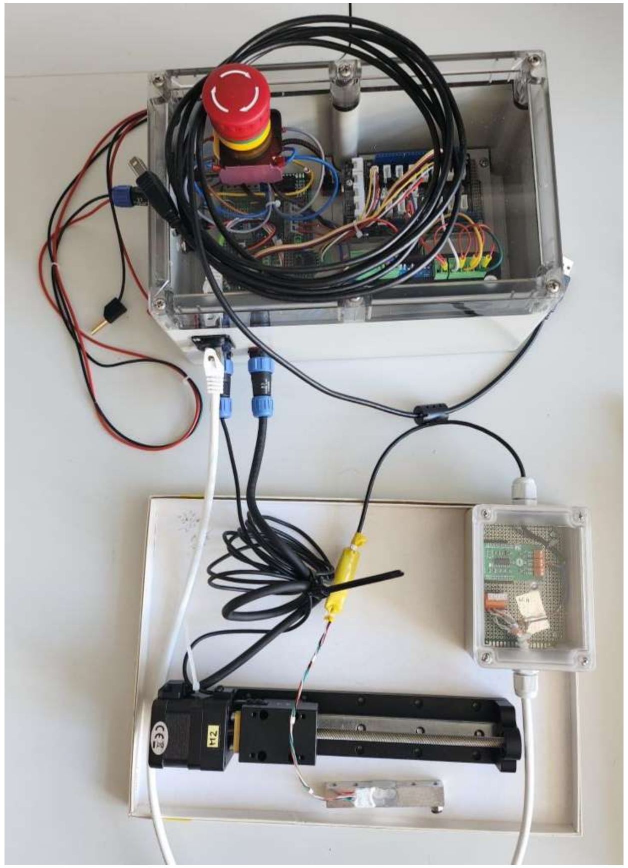
To be defined (before tests are conducted!)

Test setup

Design, construction and initial operation of test setup

- Load cell: 1kg (if required 5kg)
  - Specification, data sheet
- Linear actuator: Stroke length: ~120 mm
  - Specification, data sheet
- Arduino control unit
  - Manual instructions
- Digital microscopes
  - Manual instructions
- Access to:
  - 3D printer – Bambulab
  - Computer/Laptop



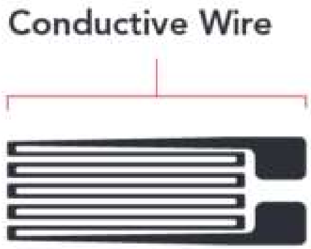


Load cell

Load cell measures weight by converting force into an electrical signal.

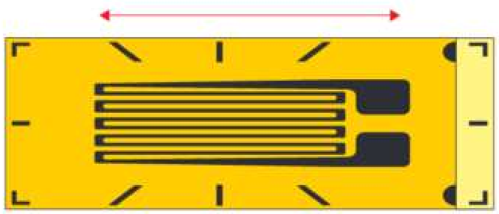
Type:

- Single-ended shear beam load cell
- Strain gauge load cell



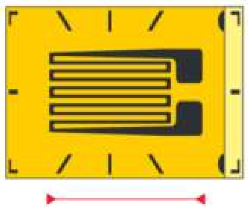
Strain gauge is an electrical element which can measure deformation of any material.

It is a very fine conductive wire or foil patterned as a grid



- When stretched → thinner → resistance increases
- When compressed → dicker → resistance decreases

Changes in resistance are very small but measurable.



Typically used in a **Wheatstone bridge** for accuracy.

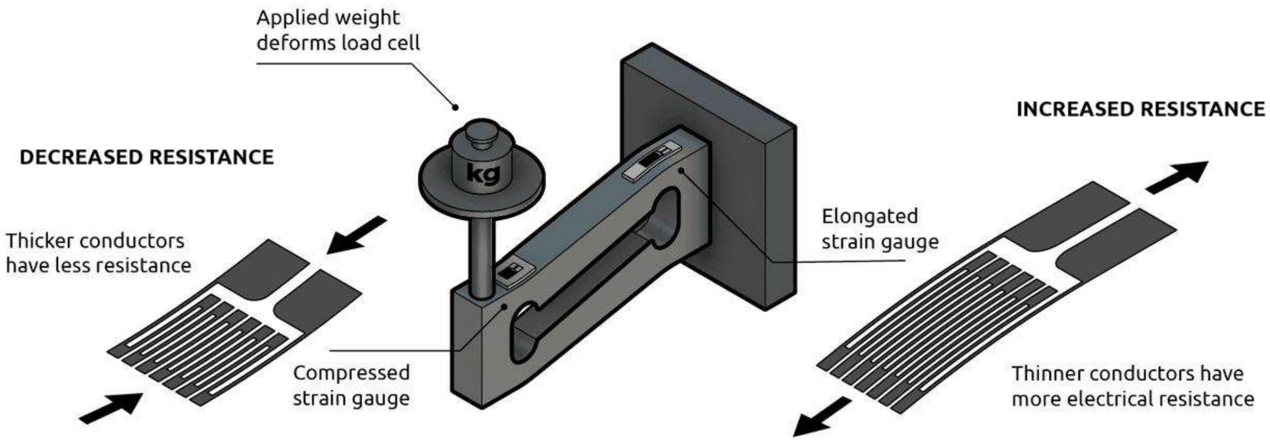
https://www.anyload.com/strain-gauge/

Load cell – Mode of operation

Strain gauges are bonded directly onto the metal surface, not visible from outside.

Usually **4 strain gauges** are used:

- 2 experience **tension**
- 2 experience **compression**



- Force applied → metal body of the load cell slightly deforms
- Strain gauges detect deformation → resistance changes

The strain gauges deform proportionally to the force applied.

- Wheatstone bridge converts resistance → small voltage signal
- Small signal → amplified, digitalized and read by a microcontroller.

https://www.flintec.com/learn/weight-sensor/load-cell

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